

Final Year Project



Moody Ping my therapist

SRS Document

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Abstract

Traditional mental health care often leaves a critical "gap" between weekly therapy sessions, leaving patients without support when they are not in a doctor's clinic. **Moody** is an AI powered ecosystem designed to bridge this gap by providing 24/7 care for patients and streamlining the workflow for therapists. The platform functions through three main pillars: an empathetic AI companion that offers instant emotional support through smart journaling and evidence-based techniques, smart AI assistants that handle administrative burdens like scheduling to let therapists focus on healing and a predictive engine that connects to real time patient data to spot early warning signs of a mental health crisis days before it happens. By combining human expertise with artificial intelligence, Moody moves mental health care from a single weekly appointment to a continuous, proactive journey, ensuring that no one must face their darkest moments alone. This document details how it must be built. It serves as the comprehensive architectural blueprint, outlining the structural design, system workflows, and database models required to bring the Moody platform to life.

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1 Introduction

The mental health system in Pakistan is currently facing a major breakdown, leaving millions of people without the help they desperately need. This crisis was recently seen in the tragic events at the **private university** [1], where students under heavy academic pressure took extreme steps because they felt they had no one to talk to in their darkest moments. In Pakistan, the problem is made worse by a massive **shortage of doctors** [2], with only one specialist available for every few hundred thousand people. Even for those who do find a therapist, there is a dangerous **"time gap"** [3] because traditional help only lasts one hour a week, leaving a person alone for the other 167 hours. On top of this, the few doctors we have are **overworked with paperwork** instead of spending time with patients, and the system is **too slow to react**, only offering help after a tragedy has already happened. Because our current system waits for a crisis rather than stopping one, it fails to protect students and ordinary people when they are most vulnerable.

1.1 Definitions

Table 1: Definitions

Term	Description
LLM	Large Language Model: An advanced artificial intelligence system trained on massive amounts of text data, enabling it to understand and generate human like text responses (used in Moody for the AI Companion)
NLP	Natural Language Processing: A branch of AI that allows computers to understand, interpret, and manipulate human language. Moody uses NLP to analyze user journal entries and auto-summarize clinical notes
CBT	Cognitive Behavioral Therapy: A widely recognized, evidence-based psychological treatment that helps individuals identify and change destructive or disturbing thought patterns. The AI companion's logic is fine-tuned on CBT principles [4].
EWS	Early Warning System: A predictive monitoring system within Moody that analyzes user data and behaviors to detect subtle signs of an impending mental health crisis before it fully develops [6].
Digital Biomarkers	Quantifiable, objective physiological and behavioral data. Examples include sleep cycles, screen time, and physical activity, which Moody uses to track mood shifts.

2 Vision Document

2.1 Problem Statement

Table 2: Problem Statement

Category	Description
Problem	• Massive shortage of mental health professionals in Pakistan, leaving millions without accessible care [3]. • A dangerous "time gap" where patients are completely unsupported between weekly 1-hour therapy sessions. • Mental health doctors are heavily burdened with administrative paperwork, reducing their capacity for actual patient care [3]. • The current healthcare system is entirely reactive, intervening only after a severe breakdown or tragedy has occurred [1]. • High social stigma prevents students and vulnerable individuals from seeking traditional help.
Affects	• Impacts vulnerable individuals, particularly students under severe academic and personal pressure. • Leaves patients feeling isolated and helpless during the 167 hours a week they are away from their therapist. • Overwhelms the limited number of available mental health professionals. • Deeply affects families, universities, and communities dealing with the fallout of preventable mental health crises.
Impact	• Tragic loss of life and severe mental health degradation. • Severe burnout among therapists and psychiatrists. • Escalation of manageable stress into severe clinical conditions. • Critical clinical time is wasted on manual scheduling, cancellations, and documentation.
Solution	• An AI-powered ecosystem that bridges the gap between weekly therapy sessions by providing 24/7 continuous support. The system must: • Offer an empathetic, always-on AI companion for instant, anonymous emotional relief and smart journaling [4]. • Provide a Clinical Assistant that automates scheduling, documentation, and prioritizes high-risk patients for therapists.

	• Utilize an Early Warning System that tracks digital biomarkers to predict and alert professionals of a crisis before it happens [6]. • Connect users with certified therapists through a secure directory while providing adaptive coping tools like mood-based music and guided breathing. • Shift the paradigm of mental health care in Pakistan from a reactive model to a proactive, continuous journey of healing.
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2.2 Business Opportunities

The **Moody** initiative provides a transformative digital opportunity within the mental health and digital therapeutics domain. By combining continuous AI support, predictive analytics, and clinical workflow automation, the system enables users to receive proactive emotional care efficiently while bridging the "**time gap**" between therapy sessions. The platform can expand into corporate wellness programs, integrate deeply with wearable tech (smartwatches) for advanced biometric tracking [5], and collaborate with national healthcare initiatives. The platform can partner with universities, private psychiatric clinics, and mental health NGOs to promote mental wellness literacy, proactive crisis intervention, and seamless access to verified therapists. It also opens opportunities for monetization through premium AI-guided coping modules, subscription-based clinical dashboards for healthcare professionals, and B2B wellness packages tailored for educational institutions.

2.3 Objectives

Following are the key objectives of Moody:

- To use AI journaling and biometrics to help users understand and manage their emotions.
- To give students a safe, anonymous way to seek help without facing social stigma.
- To automate paperwork and scheduling so doctors can focus entirely on patient healing.
- To support patients during the 167 hours a week they are away from their therapist.
- To use real-time data to predict and stop mental health emergencies before they happen.
- To offer an instant AI companion for emotional help at any time of day or night.

2.4 Scope

The scope of **Moody** encompasses the development of a cross-platform mental health and clinical support ecosystem designed to connect users with verified therapists and provide continuous AI-powered emotional assistance. The platform offers complete

functionality, including secure patient and therapist registration, authentication, smart scheduling, biometric tracking, guided journaling, and an empathetic AI companion for 24/7 support.

The initial prototype must focus on English text-based interactions and Android mobile accessibility, ensuring usability for students, general citizens, and mental health professionals across Pakistan. AI-based clinical auto-summaries and early warning systems must initially rely on digital inputs and basic mobile biometrics, while future enhancements may include regional language support (Urdu), voice-based mood analysis, deep wearable smartwatch integration for advanced health tracking, and automated crisis intervention hotlines for improved safety and inclusiveness.

2.5 Constraints

- Users providing inconsistent journal entries or irregular biometric data from their mobile devices may affect the accuracy of the early warning system and AI risk prediction.
- Requires stable internet connectivity for real-time 24/7 AI chat support, data synchronization, and automated emergency alerts.
- Availability of verified therapists and psychiatrists on the platform may limit immediate clinical connection in the early stages of deployment.
- Users lacking digital literacy or who are uncomfortable interacting with AI may face challenges navigating the self-care tools and smart journaling features.
- Health data privacy and encryption measures must strictly comply with national healthcare and digital security standards due to the highly sensitive nature of patient mental health records.

2.6 Stakeholder and User Description

2.6.1 Market Demographics

The target audience includes individuals seeking accessible emotional support, students dealing with academic or personal stress, and mental health professionals (therapists and psychiatrists) needing administrative relief and proactive patient monitoring. The primary users must be urban and semi-urban citizens, particularly university students in Pakistan, with access to smartphones or computers.

2.6.2 User Environment

Users can access Moody through a mobile application (along with a dedicated digital portal for clinicians). The interface is designed for a calming, intuitive experience, ensuring smooth navigation and consistent performance across varying devices and internet conditions to provide immediate relief during distress.

2.6.3 Stakeholder Profiles

2.6.3.1 Supervisor Team

Table 3: Supervisor Team

Representatives	Supervisor: Mr Adeel Tahir, Co-Supervisor: Mr. Tahir Farooq
Description	They are involved in supervising activities and guiding the development process.
Type	Technical stakeholders with expertise in Development and Software Engineering
Responsibility	1. Guide the development team and ensure timely completion. 2. Review documentation, prototypes, and progress. 3. Ensure that system requirements meet the academic and practical goals. 4. Provide technical validation of AI, predictive algorithms, and system features. 5. Ensure that the system aligns with intended scope and objectives.
Success Criteria	All proposed features (AI companion, clinical dashboard, early warning system) completed and validated as per project milestones.
Involvement	Review, evaluation, and continuous supervision of progress.
Comments/Issues	None

2.6.3.2 Development Team

Table 4: Development Team

Representatives	Mr. Muhammad Qasim, Mr. Shaiq Irfan, Mr. Muhammad Saif
Description	Responsible for technical implementation and documentation of the system.
Type	Technical stakeholders
Responsibility	1. Design, develop, and test all system components. 2. Document features and ensure functional reliability. 3. Implement user interface, backend, and AI modules. 4. 4. Maintain version control and testing logs.
Success Criteria	Successful completion and development of all committed features
Involvement	Documentation, development and testing of the project
Comments/Issues	None

2.6.3.3 End Users

Table 5: End Users

Representatives	General Users and Mental Health Professionals
Description	Primary users who interact with the system for continuous emotional support or managing clinical workflows
Type	External stakeholder
Responsibility	1. Use the system consistently for emotional tracking and ethical clinical practices. 2. Register, log in, use the AI companion for smart journaling, utilize coping tools, and book therapy sessions. 3. Therapists manage profiles, review AI-generated patient summaries, and manage their schedules via the clinical dashboard..
Involvement	Daily platform usage for continuous emotional support, journaling, and managing clinical workflows.
Comments/Issues	None

2.6.3.4 Admin

Table 6: Admin

Representatives	System Administrators
Description	They oversee platform operations, verify professional therapist credentials, and manage system-wide user reports.
Type	Internal Stakeholder
Responsibility	After examining the reports, accept or reject them.
Involvement	Continuous system management, verifying medical credentials and security.
Comments/Issues	None

2.6.4 Stakeholder Summary

Table 7: Stakeholder Summary

Name	Description	Responsibility
Development Team	Builds and maintains the system.	Develop, test, and document the platform.
Supervisor Team	Provides direction and guidance throughout the project.	Ensure development meets academic and system objectives.
End User	Primary beneficiaries of the platform.	Use the platform to seek continuous emotional support or manage proactive clinical care.
Admins	Manage user data and system operations.	Maintain security and performance.

3 System Requirements Specification

The Moody system is designed to provide an AI-powered mental health ecosystem that connects users with verified therapists while offering continuous 24/7 emotional support through an empathetic AI companion. It ensures seamless clinical workflow automation, secure health data tracking for early crisis prediction, and proactive access to mental health care and interactive coping tools.

3.1 System Features

The Moody system offers several core features that enhance user experience, emotional accessibility, and clinical functionality:

3.1.1 User Registration & Authentication:

The system shall allow patients, therapists, and admins to securely register and log in to the system using email and password. The system shall support password reset functionality and require a specialized verification step for mental health professionals.

3.1.2 Chat with AI Companion:

Enables real-time, empathetic, and anonymous text communication for immediate emotional relief. Users can chat with the AI at any time of day or night to navigate stress or anxiety without waiting for an appointment.

3.1.3 Write Your Thoughts:

Users can interactively log their thoughts while the system analyzes emotional patterns [4]. It also provides adaptive, proven exercises such as breathing techniques and mood-based music—tailored to how the user is feeling in the moment.

3.1.4 Early Warning and Health Tracking:

Monitors digital biomarkers (such as sleep cycles and physical activity) to predict subtle shifts in mood [6]. It automatically alerts professionals or triggers a personalized "Safety Plan" if signs of a severe mental health crisis are detected.

3.1.5 Smart Scheduling and Booking:

Allows users to search the directory and seamlessly book sessions with verified therapists. The automated booking system manages patient appointments, handles cancellations, and sends personalized reminders to reduce administrative stress.

3.1.6 Auto Summarises and Clinical Dashboard:

Acts as a digital scribe that uses NLP to convert session recordings or rough notes into structured, professional clinical summaries. It also includes an intelligent dashboard that prioritizes the therapist's daily list by flagging high-risk patient.

3.1.7 Profile Management:

Both patients and therapists can manage their personal information. Therapists can define their specializations (e.g., PTSD, Addiction, Child Psychology), ensuring clients can view accurate service details in the searchable directory.

3.1.8 Admin Dashboard:

Provides administrators with centralized control over system operations. Admins can manage user reports, verify the credentials of registered therapists, and oversee all platform health and security activities.

3.2 Functional Requirements

These requirements define the main operations that the Moody ecosystem must perform.

3.2.1 User Registration and Authentication

The system shall allow patients and therapists to create accounts and securely log in. It validates credentials, enforces health data security standards, and ensures only authorized access to personal journals or clinical dashboards.

3.2.2 Chat with AI Companion

The system shall provide a chat interface connected to a fine-tuned Natural Language Processing (NLP) model trained on evidence-based therapeutic techniques (like CBT [4]). The system shall enable users to express their feelings in natural language and receive empathetic, instant emotional support 24/7.

3.2.3 Write Your Thoughts

The system shall provide interactive mental care tools, including a guided digital diary. The system shall analyze user entries to track emotional patterns over time and recommend adaptive coping exercises, such as breathing techniques or mood-based music, based on the user's current emotional state.

3.2.4 Profile Management

The system shall enable patients and therapists to update their profiles. Patients can manage their personal preferences, while therapists can update their clinical specializations (e.g., child psychology, PTSD), contact details, and availability. All changes shall be securely stored and reflected in real time.

3.2.5 Auto Summarization

The system shall act as a digital scribe for mental health professionals. The system shall use NLP to process rough notes or session audio and automatically generate structured, professional clinical summaries to reduce the therapist's administrative burden.

3.2.6 Browse Therapist Directory

The system shall allow users to search for verified mental health professionals based on criteria such as specialization, location, or available timings. The system shall display matching therapist profiles with names, credentials, specialization tags, and available consultation slots.

3.2.7 Book Therapist Consultation

The system shall allow users to schedule consultations with verified therapists. The system shall record the booking request, automatically notify the therapist, and set the appointment status as “**Pending**” until accepted or rejected, while also sending automated reminders.

3.2.8 Patient Sorting and Early Warning Alert

The system shall monitor user mood data and digital biomarkers (such as sleep cycles or resting heart rate) [6]. The system shall automatically analyze this data to predict crisis risks and proactively flag high-risk patients on the therapist’s clinical dashboard for urgent attention.

3.2.9 Accept or Reject Consultation

The system shall allow therapists to view patient consultation requests and either accept, reject, or suggest rescheduling. The system shall update the booking status accordingly and notify the patient in real time without requiring manual administrative coordination.

3.2.10 Admin Approval

The system shall allow the admin to review and verify the medical credentials and licenses of registered therapists. The admin shall approve or reject the therapist profiles to ensure user safety. The admin shall provide feedback to the professional upon rejection.

3.3 Non-Functional Requirements

These requirements ensure the quality, performance, and usability of the Moody platform.

3.3.1 User-Friendly Interface

The system interface shall be intuitive, responsive, and accessible to all users, regardless of their technical background.

3.3.2 Data Security

All sensitive information must be encrypted using AES-256 encryption for data at rest and TLS 1.3 for data in transit. The platform shall implement standard token-based authentication with a maximum session timeout of 30 minutes for the clinical dashboard to prevent unauthorized access.

3.3.3 Performance

The system shall maintain a maximum API response time of ≤ 2 seconds for 95% of requests under a load of up to 500 concurrent users. The fine-tuned AI companion must return responses within ≤ 5 seconds.

3.3.4 Privacy

The system shall enforce strict Role-Based Access Control (RBAC), ensuring patients can only access their own profiles and therapists can only view explicitly assigned patient records.

3.3.5 Compatibility

The patient facing mobile application shall be fully supported and operate without UI degradation on Android 10+ operating systems.

4 Use Case Diagram:



Figure 1: Use Case Diagram

5 High-Level Use Cases

5.1 Registration

Table 8: Registration High Level Use Case

Use Case ID	UC-01
Use Case Name	Registration
Type	Primary
Actor(s)	User, Therapist
Description	Allows users and Therapist to create a new account or sign in using valid credentials.
Preconditions	The user or therapist must have an internet connection, and the application must be running.
Postconditions	The actor is authenticated and redirected to their respective dashboard (User/Patient interface or Clinical Dashboard).

5.2 Update Profile

Table 9: Update Profile High Level Use Case

Use Case ID	UC-02
Use Case Name	Update Profile
Type	Secondary
Actor(s)	User, Therapist
Description	Allows users and Therapist to edit and update their personal or professional details including name, contact information, and specialization.
Actor(s)	User, Therapist
Preconditions	Actor must be logged in.
Postconditions	Updated profile details are saved and reflected in the system.

5.3 Chat with AI-Companion

Table 10: Chat with AI-Companion High Level Use Case

Use Case ID	UC-03
Use Case Name	Chat with AI-Companion
Type	Primary
Actor(s)	User
Description	Provides a real-time, 24/7 communication channel between users and the fine-tuned empathetic AI model. Users can share their emotional state, discuss stressors, or seek immediate support, and the model responds with empathetic conversation and evidence-based coping strategies.
Preconditions	Actor must be logged in and have access to internet.
Postconditions	Actor receives a relevant, empathetic response to their emotional state or query.

5.4 Auto-Summarize Clinical Notes

Table 11: Auto-Summarize Clinical Notes High Level Use Case

Use Case ID	UC-04
Use Case Name	Auto-Summarize Clinical Notes
Type	Primary
Actor(s)	Therapist
Description	Allows therapists to upload session recordings or input rough clinical notes. The system uses Natural Language Processing (NLP) to extract key insights and automatically generates a professional, structured clinical summary, significantly reducing the therapist's administrative burden
Preconditions	Actor must be logged in; supported file type (Audio, PDF, DOC, Text).
Postconditions	A professional, structured clinical summary is generated and displayed to the actor.

5.5 Access Coping Tools

Table 12: Access Coping Tools High Level Use Case

Use Case ID	UC-05
Use Case Name	Access Coping Tools
Type	Primary
Actor(s)	User
Description	Displays interactive, step-by-step coping mechanisms and evidence-based exercises for managing stress and anxiety. Users can view and utilize tailored tools such as guided breathing techniques, mood-based music, and smart journaling prompts that adapt to their current emotional state.
Preconditions	Actor logged in and selected a specific Module.
Postconditions	Actor successfully views or participates in the full exercise.

5.6 Offer Services

Table 13: Offer Services High Level Use Case

Use Case ID	UC-06
Use Case Name	Offer Services
Type	Primary
Actor(s)	Therapist
Description	Enables therapists to list, modify, or remove specific clinical services and specializations they provide, such as Child Psychology, PTSD treatment, or Addiction/Recovery. The use case includes credential and medical license verification by the admin before the profile and services become visible to users.
Preconditions	Therapists must be logged in and verified before offering services. Users must be signed in to search or book therapists. Admin must have access rights to review service listings.
Postconditions	Verified services appear on the therapist's public profile, available for user viewing and booking. Booking status updates (Accepted/Rejected) are reflected in real time.

5.7 Browse Therapist Directory

Table 14: Browse Therapist Directory High Level Use Case

Use Case ID	UC-07
Use Case Name	Browse Therapist Directory
Type	Primary
Actor(s)	User
Description	Allows users to search the directory for certified mental health professionals based on criteria such as specialization, location, and available timings. Once a suitable therapist is found, users can seamlessly initiate a consultation booking.
Preconditions	The user must be logged in and have an active internet connection. The system must contain verified Therapist profiles.
Postconditions	A consultation booking request is successfully created with a “Pending” status, and the Therapist is notified in real-time.

6 Expanded Use Cases

6.1 Registration

Table 15: Sign Up/Sign In Expanded Use Case

Use Case ID	UC-01
Use Case Name	Registration
Description	Enables users and therapists to register for a new account or sign in to access the Moody platform. This establishes secure entry and ensures that only verified individuals can access sensitive emotional data and clinical services.
Actor(s)	User, Therapist
Preconditions	The user or therapist must have an internet connection, and the application must be running.
Postconditions	The actor is authenticated and redirected to their respective dashboard (User/Patient interface or Clinical Dashboard).
Normal Flow	1. Actor opens the Moody application. 2. Selects role as “User” or “Therapist”. 3. Enters email/phone and password or selects “Sign Up” and completes registration. 4. System validates credentials through the Authentication process. 5. The system verifies user identity, matches credentials with the secure database, and generates a session token. 6. On success, the dashboard loads according to the actor’s role.

Alternative Flow	3.1. If the user clicks “Forgot Password”, the system prompts the user to enter their registered email address. 3.2. The user enters their email, the system sends email containing a password resent link to user. 3.3. The user clicks the link and resets the password and logs in. 4.1. If the authentication fails, the user is redirected to sign up page. 4.2. If the email is already registered, the system shows “Email already exists”. 4.3. If user enters invalid credentials, the system displays “Invalid Email/Password” and redirects back to the Sign Up page.
Special Requirements	1. Passwords must be 8 characters long and include at least one uppercase letter, lowercase, number, special character, with no spaces or common words 2. Requires a stable internet connection.
Assumptions	Users and Therapist provide valid contact information and unique credentials. The authentication module maintains a real-time connection with the user database.
Frequency of Use	High

6.2 Update Profile

Table 16: Update Profile Expanded Use Case

Use Case ID	UC-02
Use Case Name	Update Profile
Description	Allows users and Therapist to edit their personal or professional details, such as phone number, address, or specialization, ensuring their information remains current.
Actor(s)	User, Therapist
Preconditions	Actor must be logged in.
Postconditions	Updated profile details are saved and reflected in the system.
Normal Flow	1. Actor opens “Profile” section. 2. Clicks “Edit Profile.” 3. Updates fields such as phone number, address, or specialization. 4. Clicks “Save Changes.” 5. System validates inputs and updates the database.
Alternative Flow	3.1. If any field is empty, the system shows “Please complete all required fields”. 4.2. If database update fails, the system shows “Unable to save changes”. 4.3. If unauthorized edits occur, the system shows “Permission denied”.
Special Requirements	Requires a stable internet connection.
Assumptions	Database connectivity is stable during profile updates.

Frequency of Use	Medium
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6.3 Chat with AI-Companion

Table 17: Chat with AI-Companion Expanded Use Case

Use Case ID	UC-03
Use Case Name	Chat with AI-Companion
Description	Facilitates text-based, empathetic communication between users and the fine-tuned LLM model for immediate emotional support. The model provides simplified, evidence-based coping strategies (like CBT techniques).
Actor(s)	User
Preconditions	Actor must be logged in and have access to internet.
Postconditions	Actor receives a relevant, empathetic response to their emotional state or query.
Normal Flow	1. Actor opens "AI Companion." 2. Enters a thought, feeling, or question in natural language. 3. System sends the input to the AI engine. 4. AI Engine analyzes the emotional context and generates an empathetic, supportive response. 5. System displays the response in the chat window.
Alternative Flow	3.1. If AI model is unavailable, show "Service temporarily offline". 4.1. If the system detects severe self-harm intent, it triggers the "Emergency Crisis/Safety Plan" overlay. 5.1. If connection is lost, show "Please retry".
Special Requirements	1. LLM model must be fine-tuned on verified psychological first-aid and therapeutic frameworks. 2. Requires a stable internet connection. 3. Strict data privacy and end-to-end encryption for all conversations.
Assumptions	The AI Engine can accurately detect sentiment and provide safe, non-diagnostic support.
Frequency of Use	High

6.4 Auto Summarize Clinical Notes

Table 18: Auto Summarize Clinical Notes Expanded Use Case

Use Case ID	UC-04
Use Case Name	Auto-Summarize Clinical Notes
Description	Allows therapists to upload audio recordings or rough text notes for AI-based clinical summarization and structuring.
Actor(s)	Therapist
Preconditions	Actor must be logged in; supported file type (Audio, PDF, DOC, Text).
Postconditions	A professional, structured clinical summary is generated and displayed to the actor.
Normal Flow	1. Actor opens “Clinical Dashboard” and selects a patient profile. 2. Selects "Generate Summary" and uploads a file or inputs rough notes. 3. System uploads the file and runs Speech-to-Text if it is an audio file. 4. AI Engine processes the text, extracts key clinical insights, and structures it. 5. System displays the simplified clinical summary to the actor for review.
Alternative Flow	1.1. If file is corrupted or missing, show “Invalid file”. 3.1. If format is unsupported, display “Upload only Audio/PDF/DOCX”. 4.1. If processing fails, show “Error during summarization”.
Special Requirements	1. Uploaded files/audio are not stored permanently; all text processing is highly secure and complies with health data privacy laws. 2. NLP support for medical/psychological terminology. 3. Requires a stable internet connection.
Assumptions	AI Engine correctly interprets clinical context and therapeutic terminology.
Frequency of Use	High

6.5 Access Coping Tools

Table 19: Access Coping Tools Expanded Use Case

Use Case ID	UC-05
Use Case Name	Access Coping Tools
Description	Provides structured, interactive guides for mental wellness exercises such as guided breathing, grounding techniques, or anxiety management.
Actor(s)	User
Preconditions	Actor logged in and selected a specific Module.
Postconditions	Actor successfully views or participates in the full exercise.
Normal Flow	1. Actor opens “Coping Tools.” 2. Selects a category (e.g., Panic Relief, Sleep Aid, Focus). 3. System retrieves the exercise details from the database. 4. Steps, timers, and accompanying audio/visuals are displayed interactively.
Alternative Flow	1.1. If selected module is missing, system shows “Exercise unavailable”. 3.1. If database fetch fails, system shows “Unable to load details”.
Special Requirements	1. Data must be clinically verified and sourced from certified mental health professionals. 2. Exercises shall be responsive and adaptive to user inputs. 3. Requires a stable internet connection.
Assumptions	Coping mechanisms follow standard, evidence-based psychological practices.
Frequency of Use	Medium

6.6 Therapy Service Management

Table 20: Therapist Services Expanded Use Case

Use Case ID	UC-06
Use Case Name	Therapy Services Management
Description	The system enables therapists to offer, manage, and verify clinical services while allowing users to search for and book consultations. Admins review and approve therapist-provided services and licenses before they become visible to users, ensuring medical authenticity and professional credibility.
Actor(s)	Therapist, User, Admin
Preconditions	Therapists must be logged in and verified before offering services. Users must be signed in to search or book therapists. Admin must have access rights to review service listings.
Postconditions	Verified services appear on the therapist’s public profile, available for user viewing and booking. Booking status updates (Accepted/Rejected) are reflected in real time.

Normal Flow	1. Therapist logs into the platform and opens “My Services.” 2. Adds new service details such as title, specialization (e.g., PTSD, Child Psych), and description. 3. Submits the service/credentials for review and verification. 4. The system saves the service entry and marks it as “Pending.” 5. The admin accesses “Pending Services,” reviews therapist medical licenses and details. 6. The admin approves or rejects the service. 7. Upon approval, the service becomes visible in the therapist’s public profile. 8. Users open “Find Therapist” and enter search filters. 9. The system retrieves matching verified therapist profiles. 10. Users view details and select a therapist for consultation. 11. Users submit a booking request with preferred time. 12. Therapists open “Consultation Requests,” review pending bookings, and choose to “Accept” or “Reject.” 13. The system updates booking status and sends notifications to both parties.
Alternative Flow	1.1. If mandatory service fields are incomplete, system prompts “Please complete all details.” 1.2. If a duplicate service is detected, system displays “Service already listed.” 5.1. If therapist credentials are incomplete, admin marks the record as “Verification Required.” 6.1. If inconsistencies are found, admin rejects the service and notifies the therapist. 8.1. If filters are empty, system prompts “Enter valid search criteria.” 8.2. If no therapists match, system displays “No therapists found.” 12.1. If there is a network issue, system displays “Unable to fetch requests.” 12.2. If the update fails, system shows “Request unsuccessful, please retry.”
Special Requirements	1. Admin verification of medical credentials is mandatory before services are published. 2. Real-time synchronization must occur for booking status across user and therapist dashboards. 3. Search results must only display verified professionals.
Assumptions	Therapists provide accurate specialization details and respond promptly. Admins act strictly according to healthcare verification standards.
Frequency of Use	High

6.7 Browse Therapist Directory

Table 21: Browse Therapist Directory Expanded Use Case

Use Case ID	UC-07
Use Case Name	Browse Therapist Directory
Description	Enables users to search for verified mental health professionals based on specialization, location, or availability, and book consultations. The system provides verified profiles with contact information, ratings, and available time slots.
Actor(s)	User
Preconditions	The user must be logged in and have an active internet connection. The system must contain verified Therapist profiles.
Postconditions	A consultation booking request is successfully created with a “Pending” status, and the Therapist is notified in real-time.
Normal Flow	1. User opens the “Find Therapist” feature. 2. Enters search filters such as specialization (e.g., Anxiety, Depression), city, or availability. 3. System retrieves and displays a list of verified therapists matching the criteria. 4. User views profiles with details like credentials, experience, and consultation fees. 5. User selects a preferred therapist and chooses “Book Consultation.” 6. User enters consultation details, including preferred date and time. 7. System validates the request and creates a booking entry with “Pending” status. 8. The system sends a real-time notification to the selected therapist for confirmation.
Alternative Flow	2.1. If filters are empty, system shows “Enter valid search criteria”. 2.2. If no match is found, system shows “No therapists found”. 6.1. If the selected time slot is unavailable, the system displays “Time not available.” 7.1. If booking creation fails, the system shows “Unable to process request.”
Special Requirements	1. Only verified therapists shall be displayed in the search results. 2. A real-time notification mechanism must update booking statuses instantly. 3. The system must maintain secure, private communication between users and therapists.
Assumptions	Therapists are actively checking their dashboards for new bookings. Users provide accurate booking details.
Frequency of Use	High

7 Activity Diagram

7.1 Activity Diagram of User

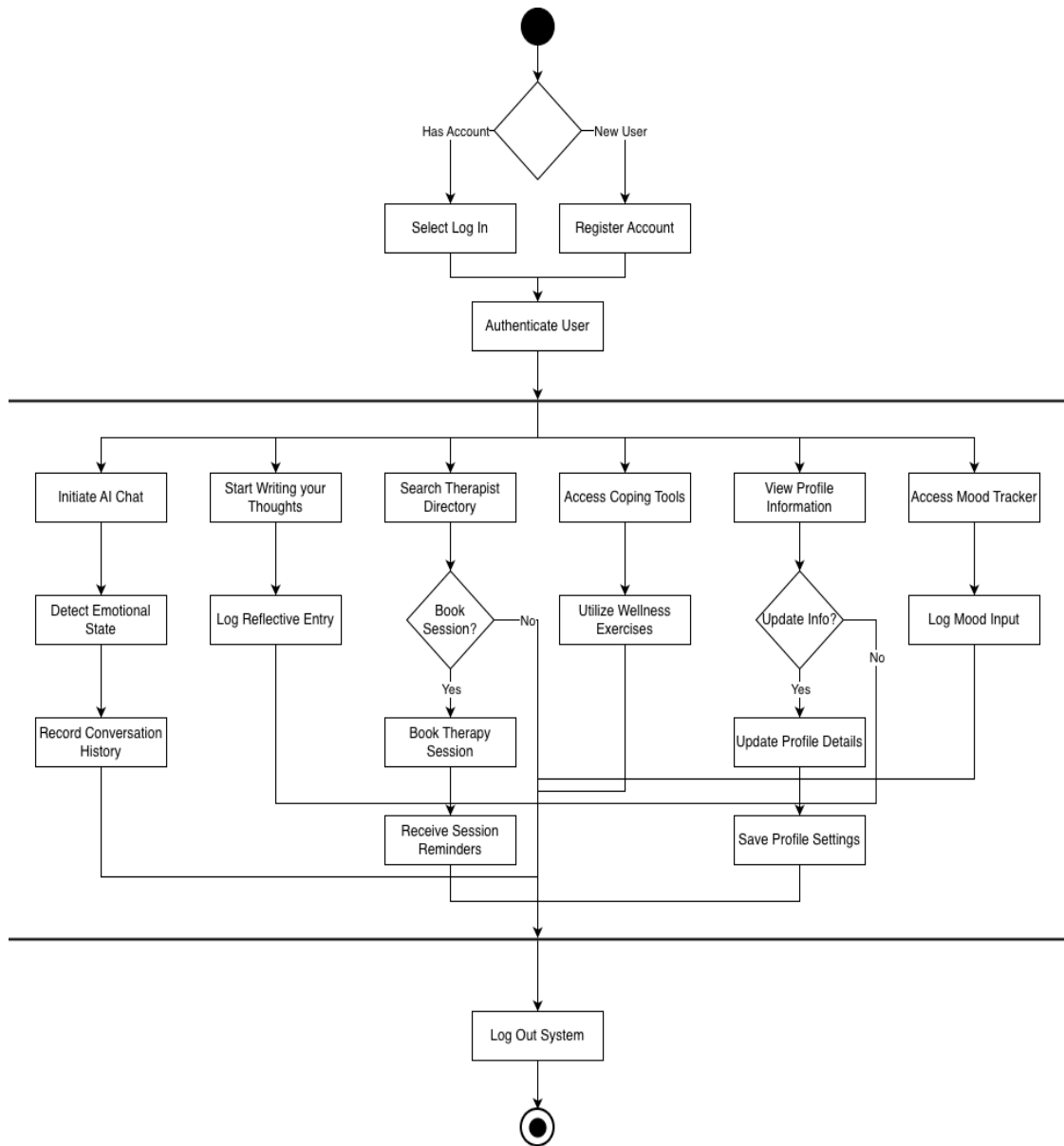


Figure 2: User Activity Diagram

7.2 Activity Diagram of Therapist

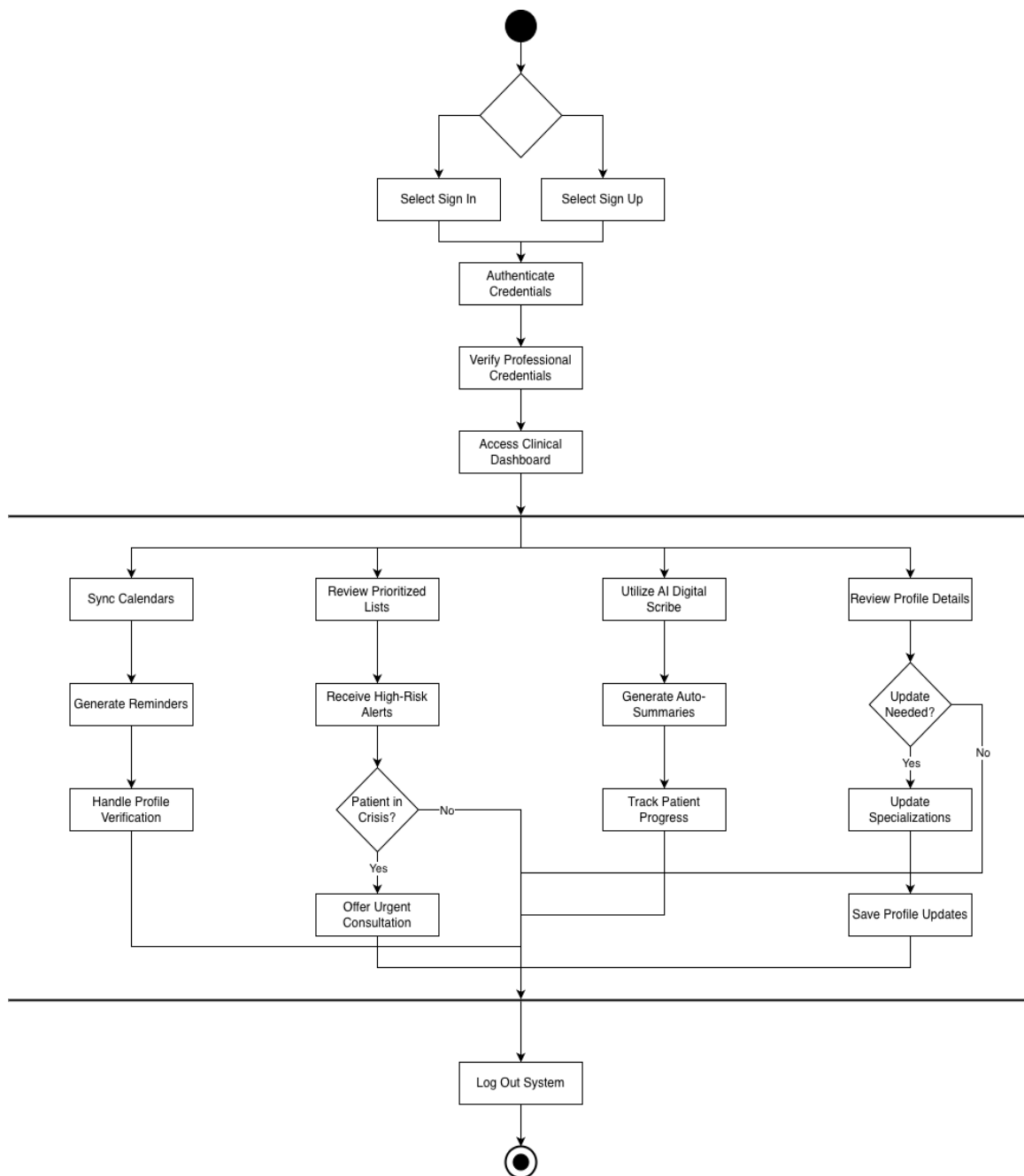


Figure 3: Therapist Activity Diagram

7.3 Activity Diagram of Admin

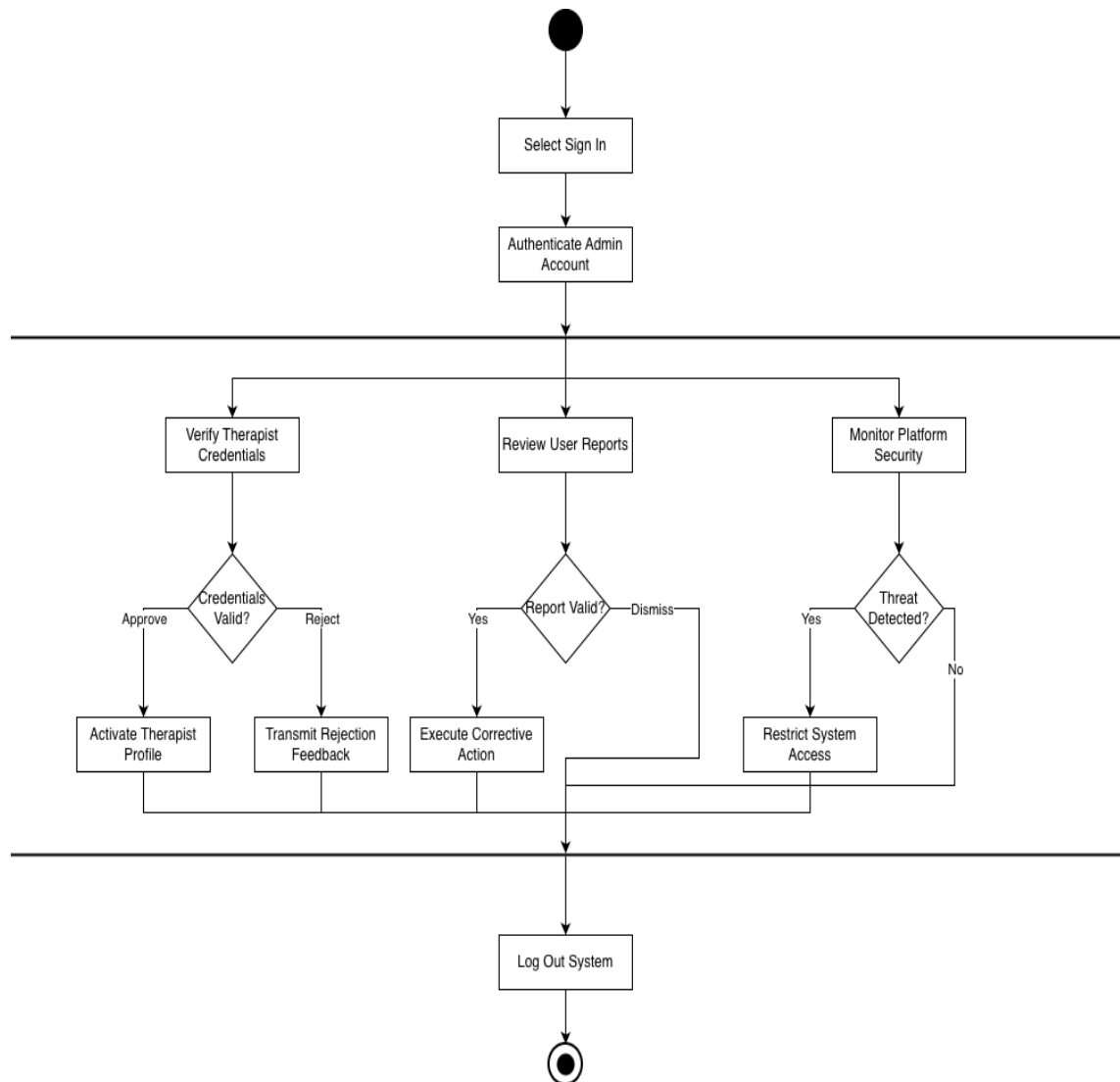


Figure 4: Admin Activity Diagram

8 System Sequence Diagram

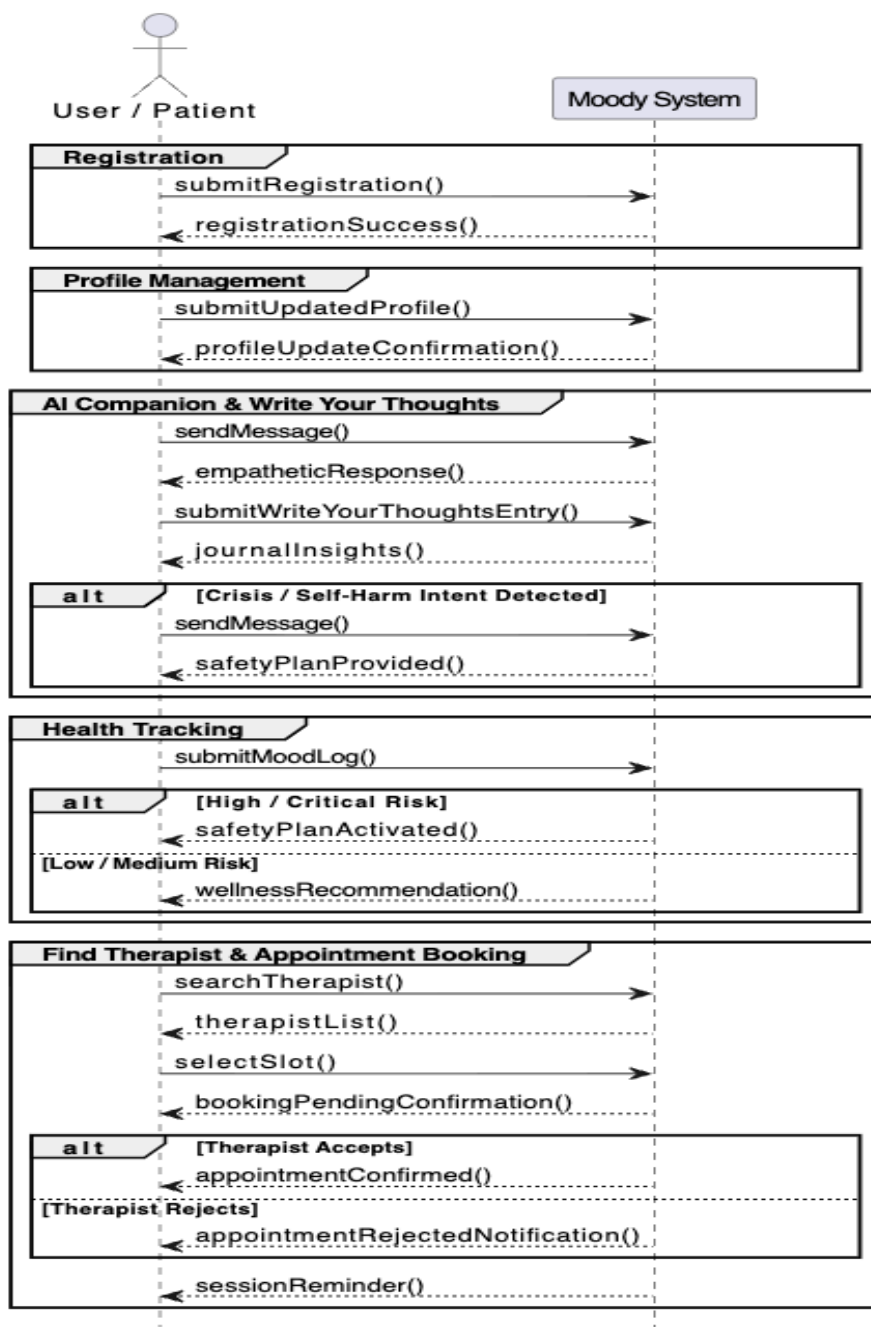
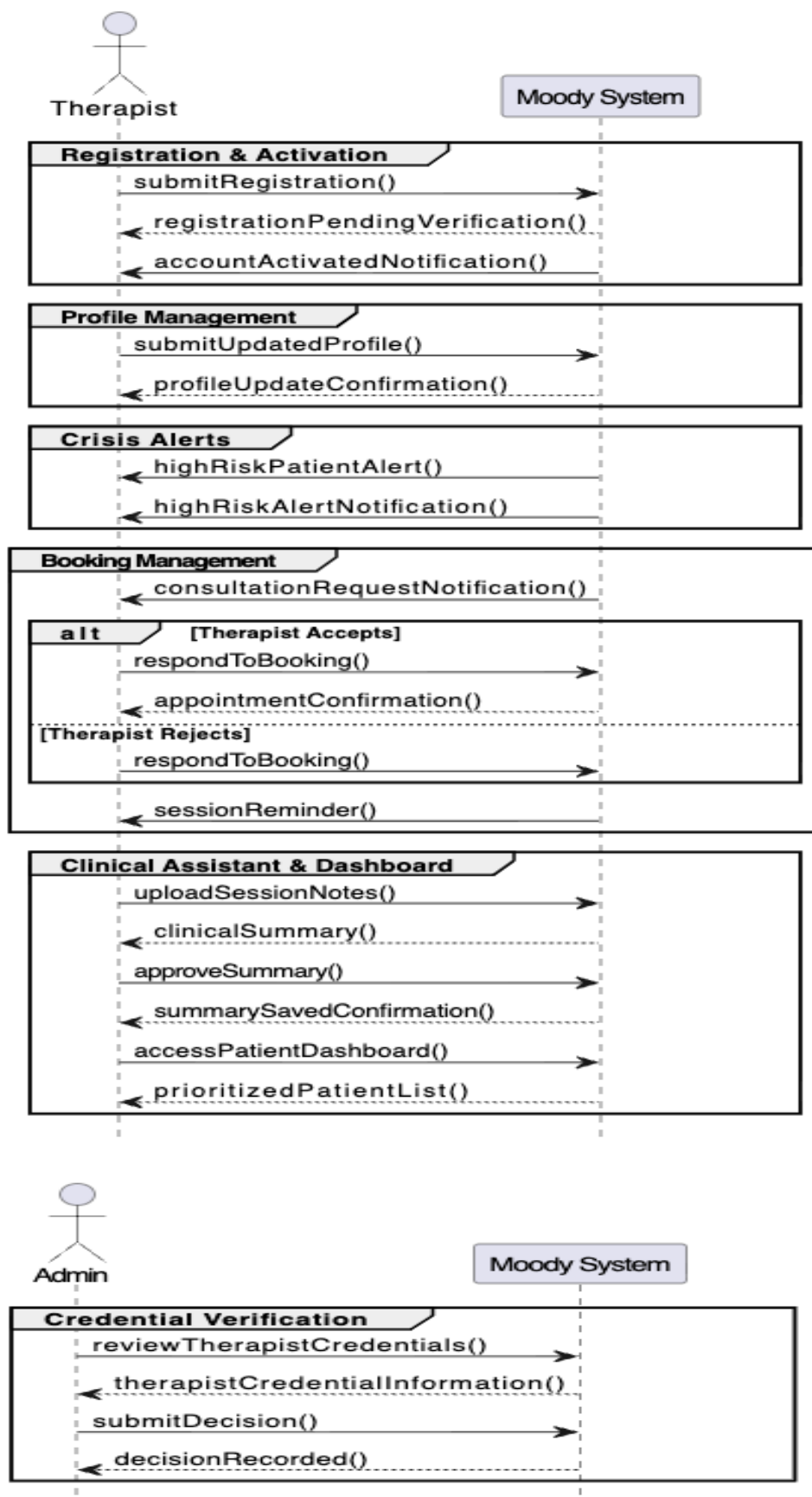


Figure 5: System Sequence Diagram

Moody Ping my Therapist



9 Sequence Diagrams

9.1 Chat with AI-Companion

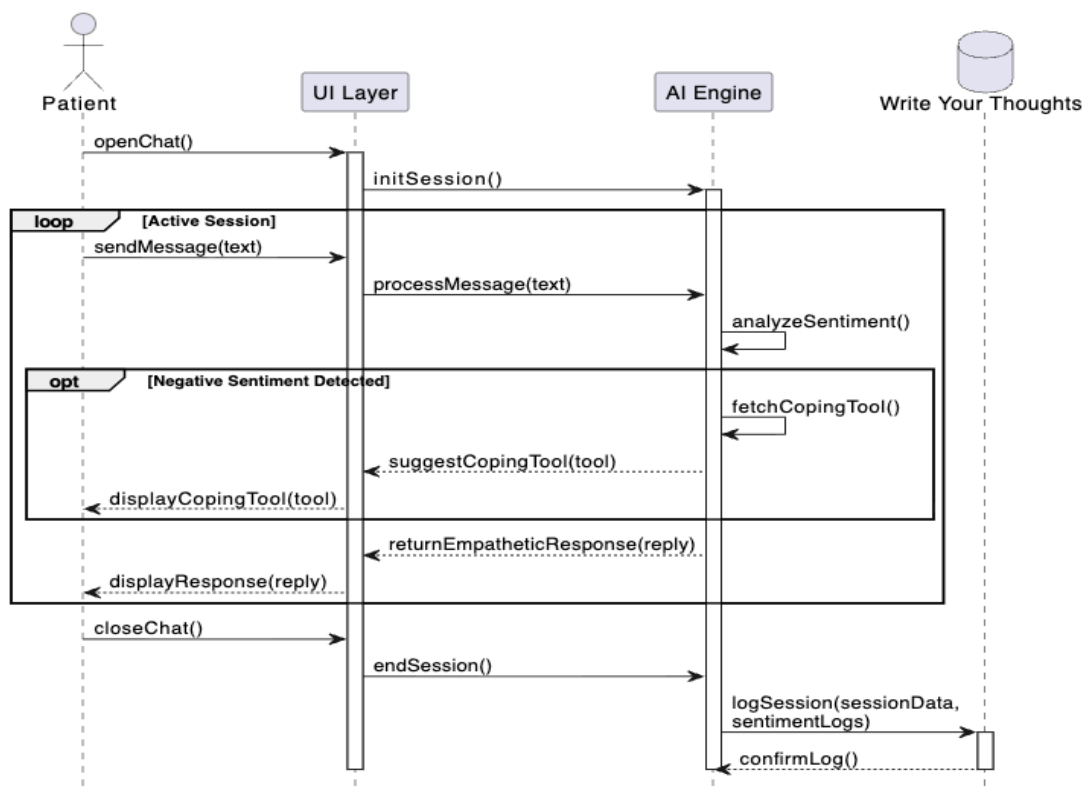


Figure 6: Chat With AI Companion

9.2 Early Warning System

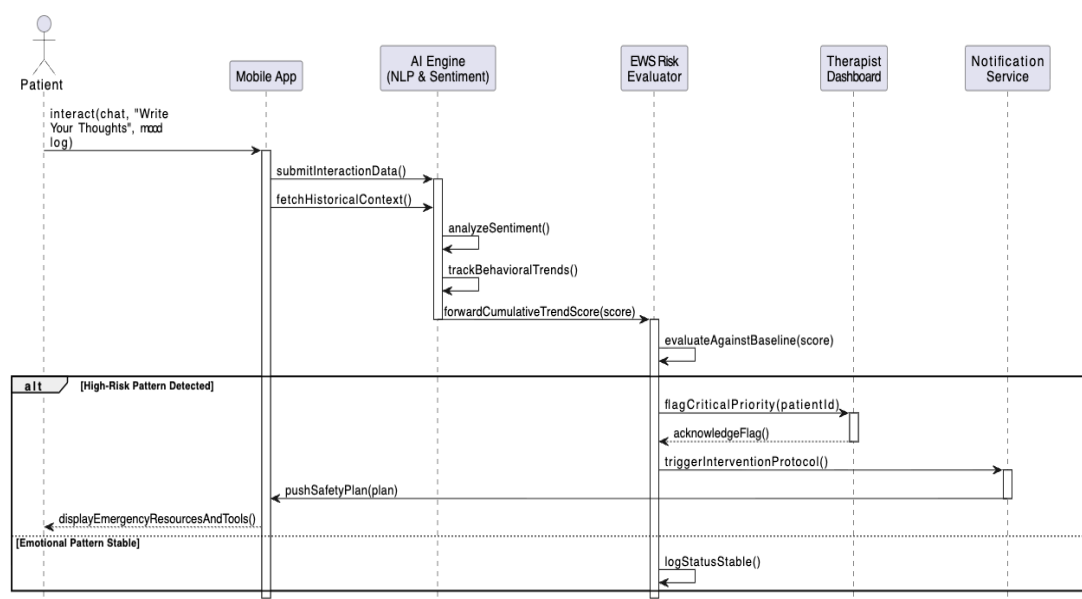


Figure 7: Early Warning System

9.3 Clinical Assistant

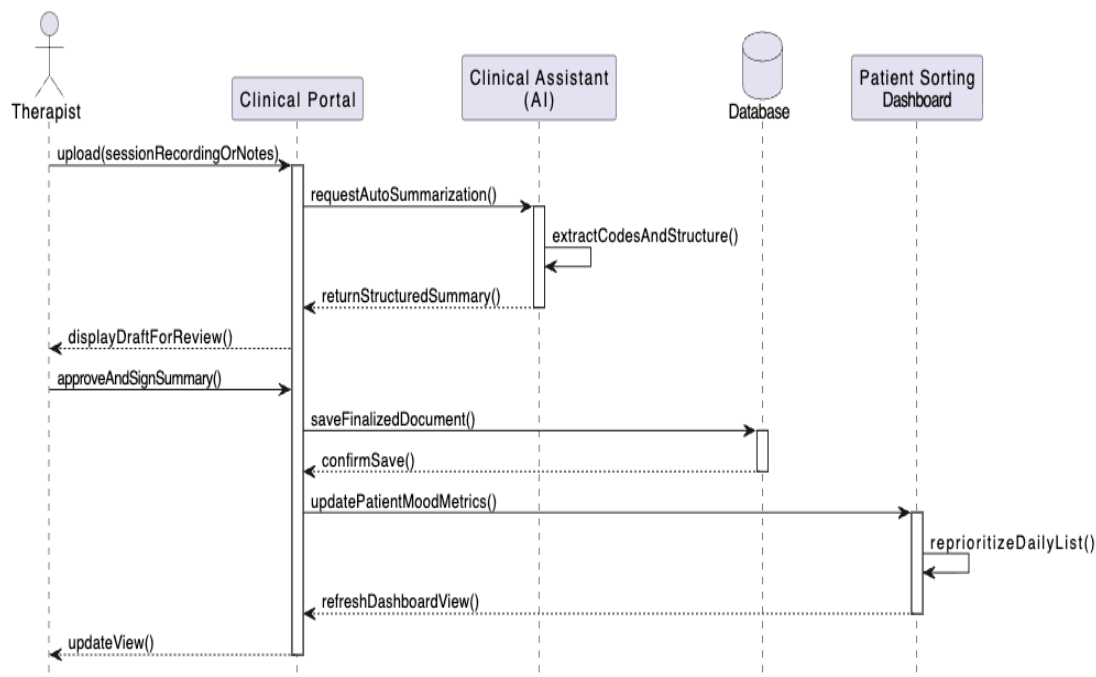


Figure 8: Clinical Assistant

9.4 Browse Therapist Directory

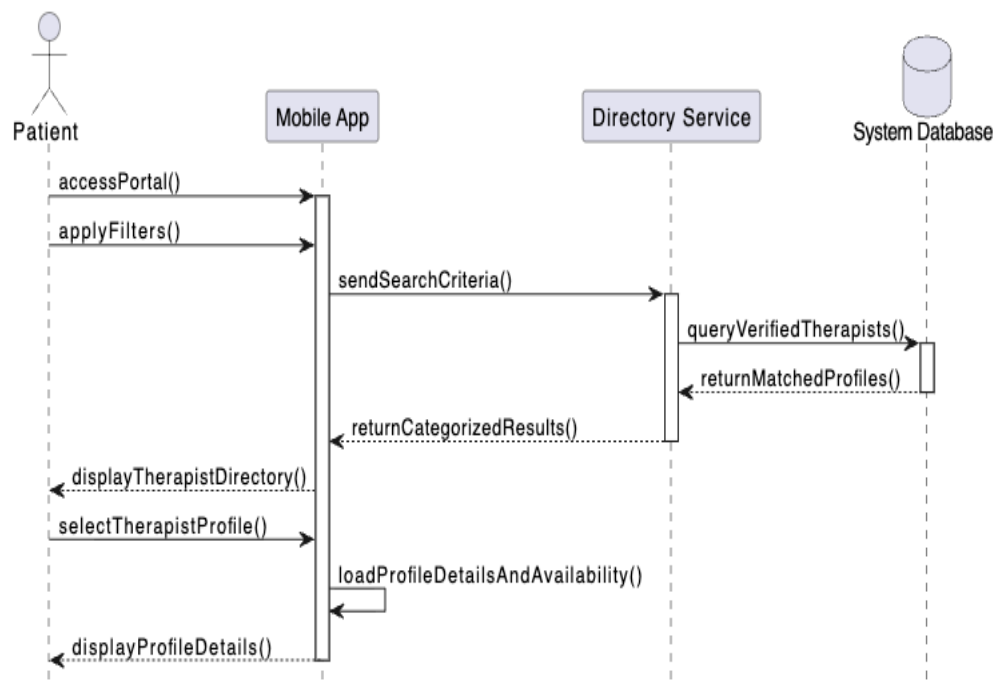


Figure 9: Browse Therapist Directory

9.5 Therapist Registration and Verification Flow

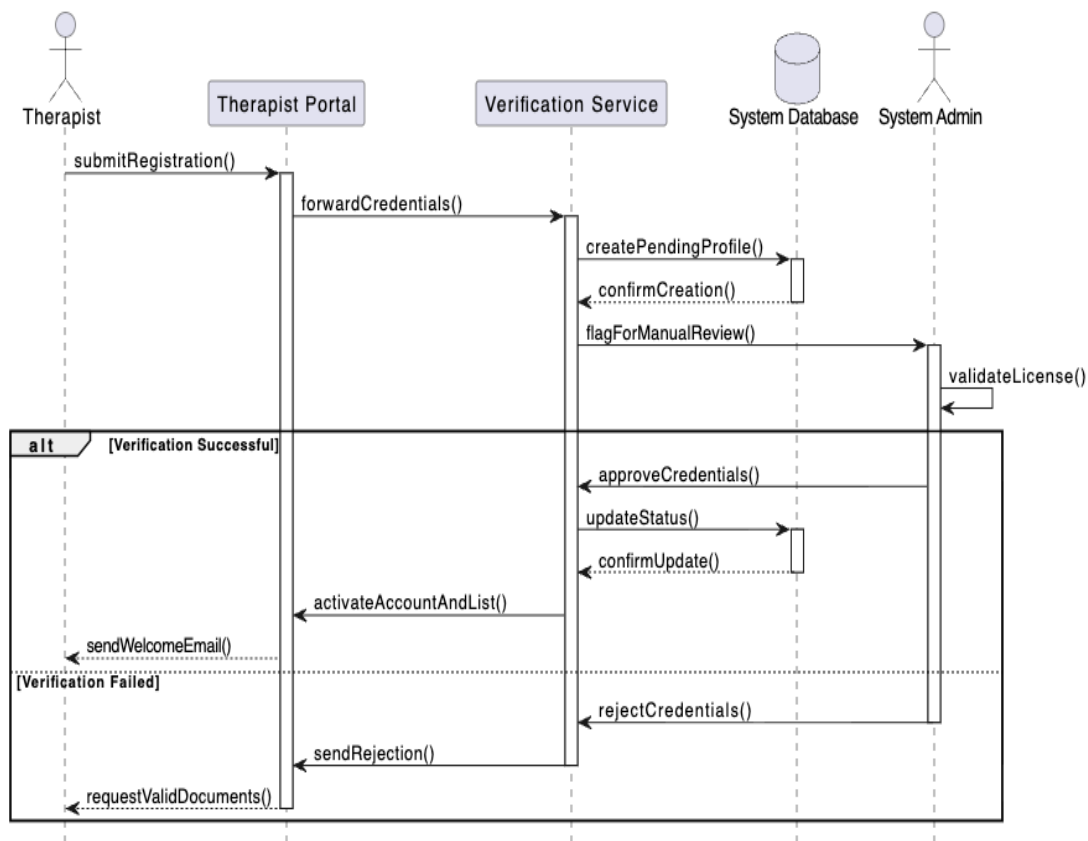


Figure 10: Therapist Registration System Sequence Diagram

9.6 Smart Appointment Scheduling and Reminders

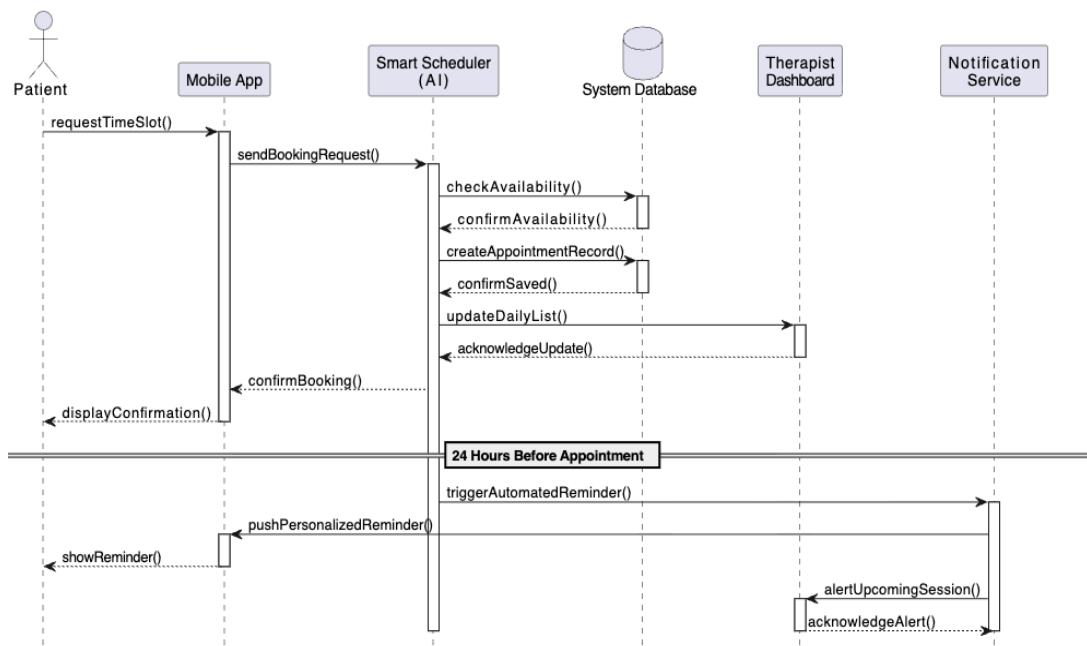


Figure 11: Smart Appointment Scheduling

9.7 Write Your Thoughts

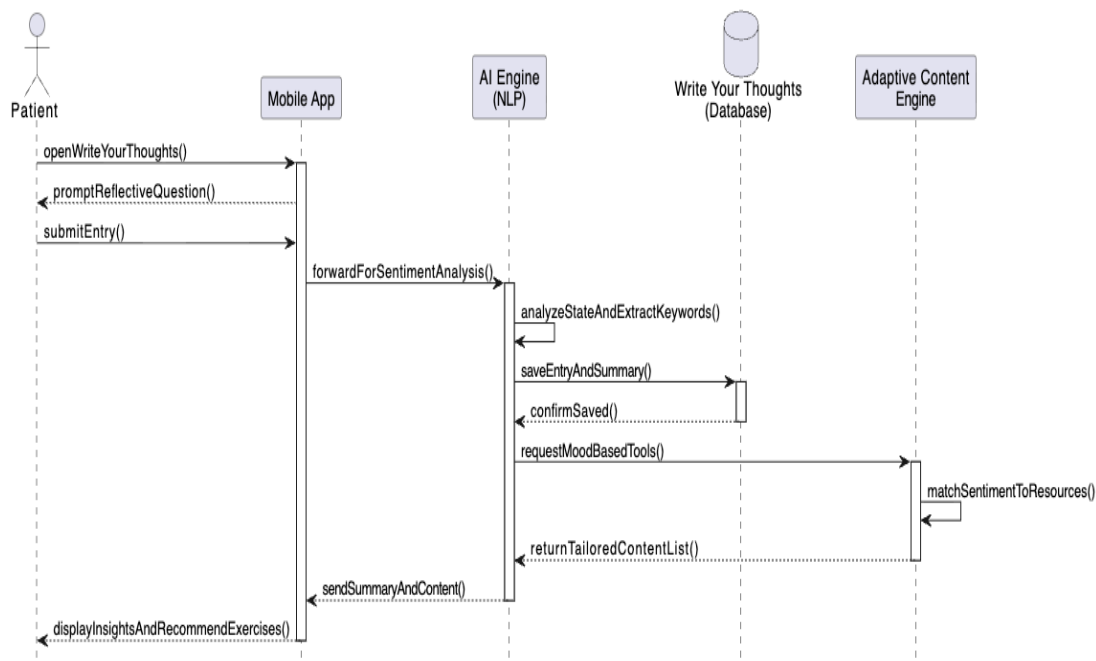


Figure 12: Write Your Thoughts

9.8 Patient Sorting and Prioritization Flow

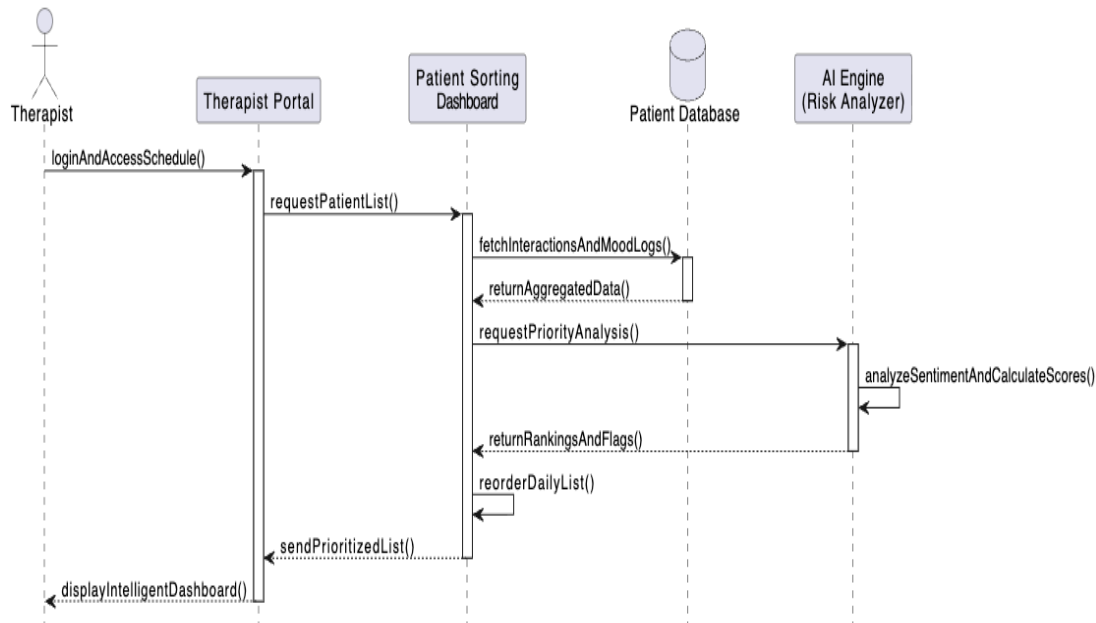


Figure 13: Patient Sorting

9.9 Automated Emergency “Safety Plan” Execution

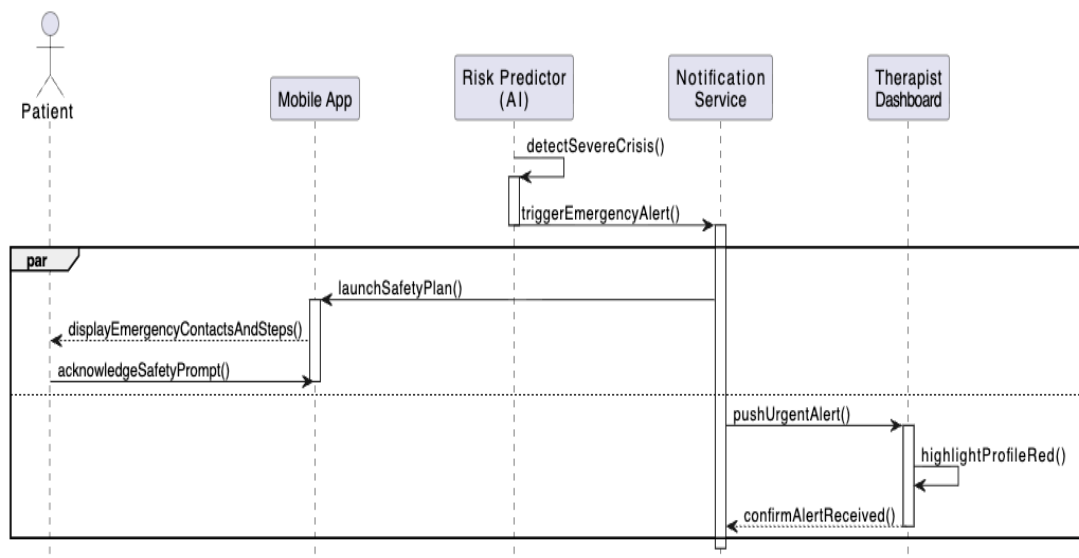


Figure 14: Safety Plan

10 Data Flow Diagram

10.1 DFD Level 0

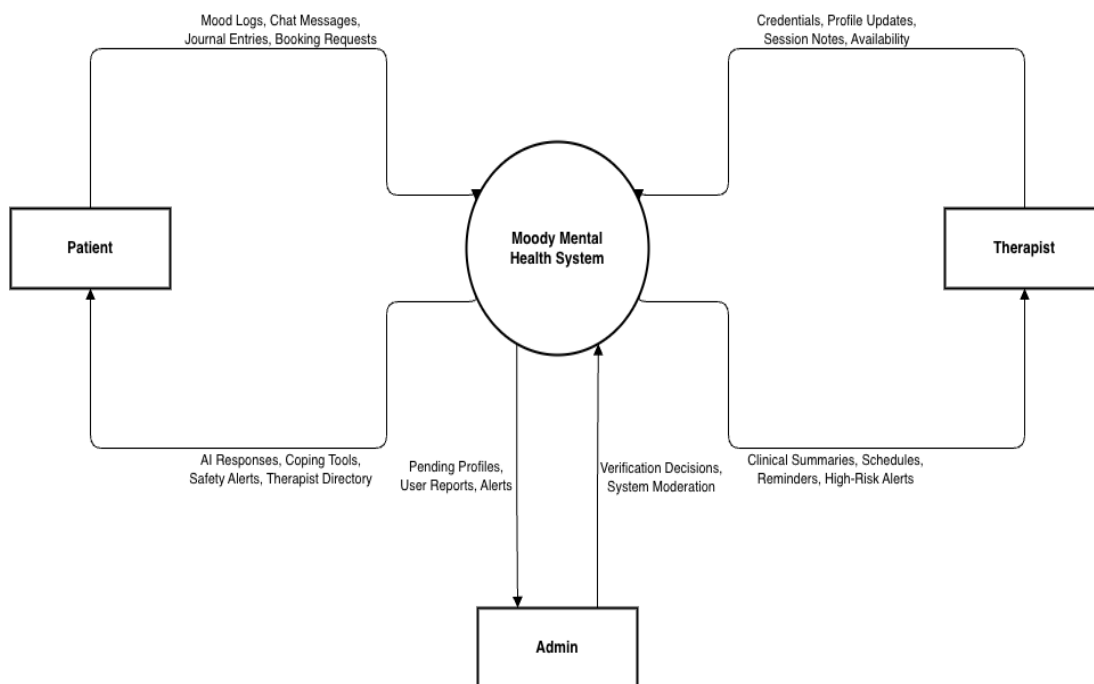


Figure 15: DFD Level 0

10.2 DFD Level 1

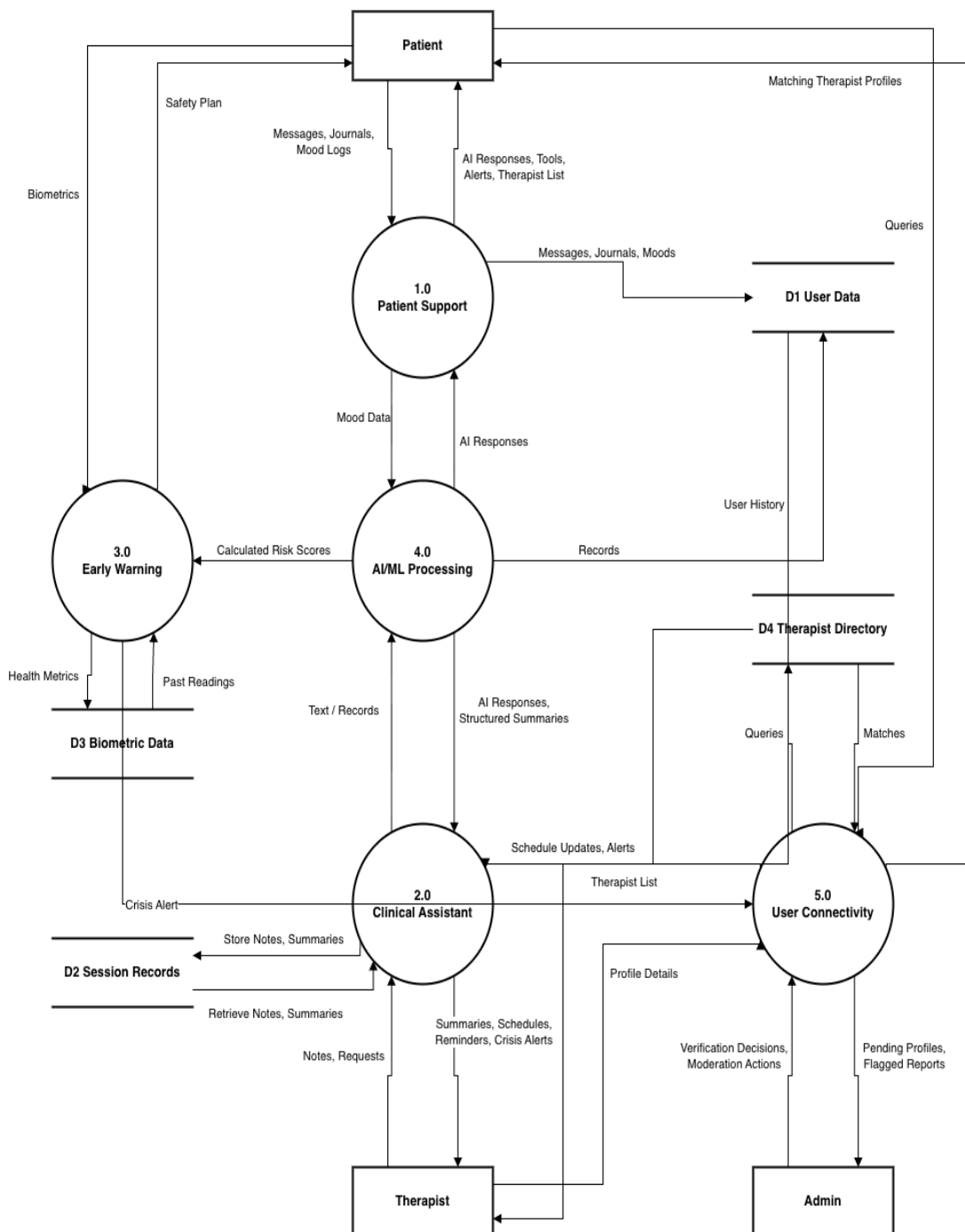
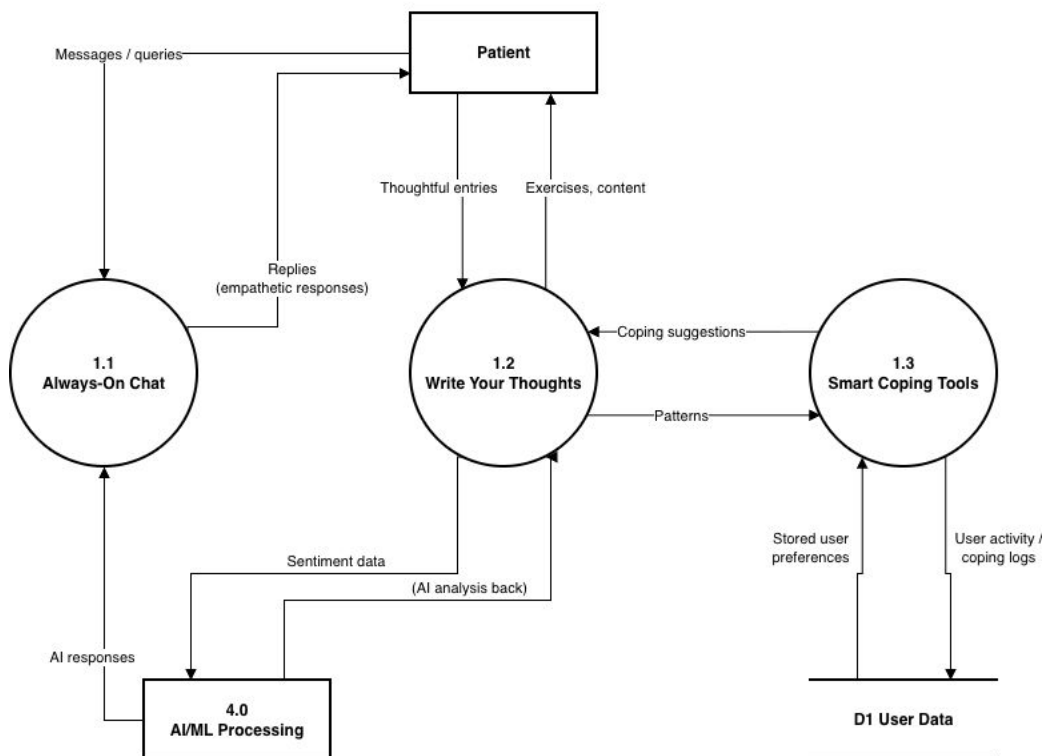


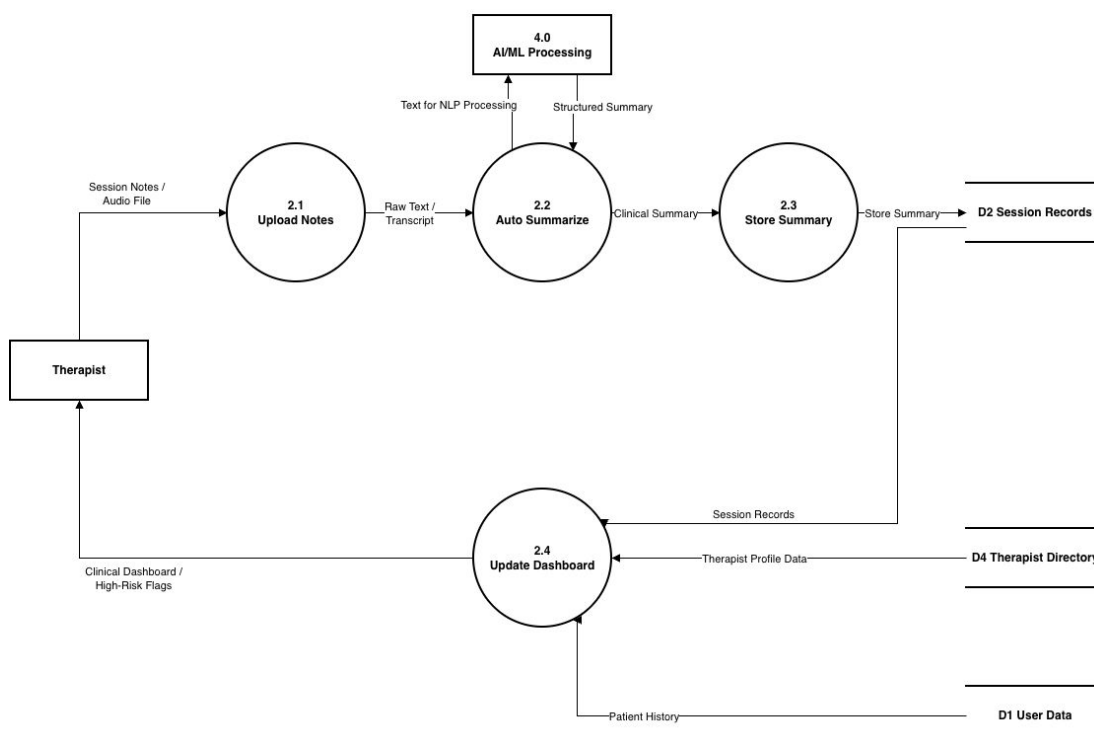
Figure 16: DFD Level 1

10.3 DFD Level 2

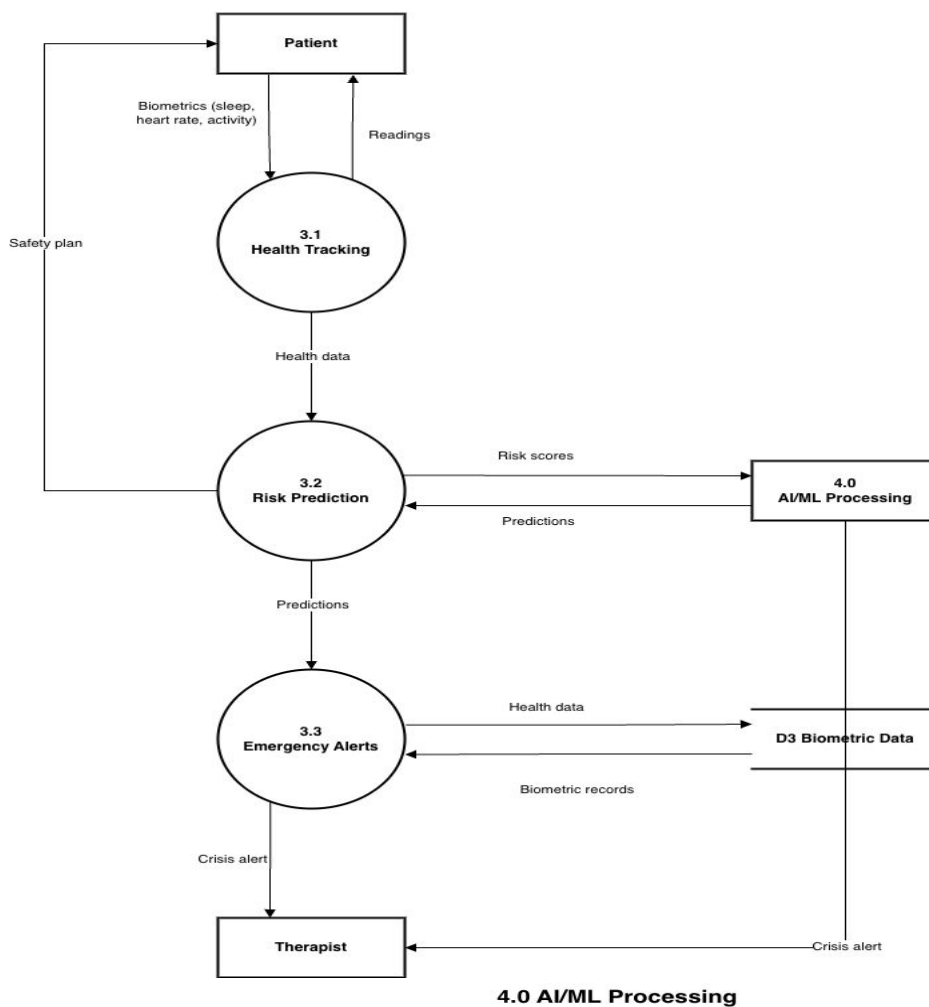
1.0 Patient Support



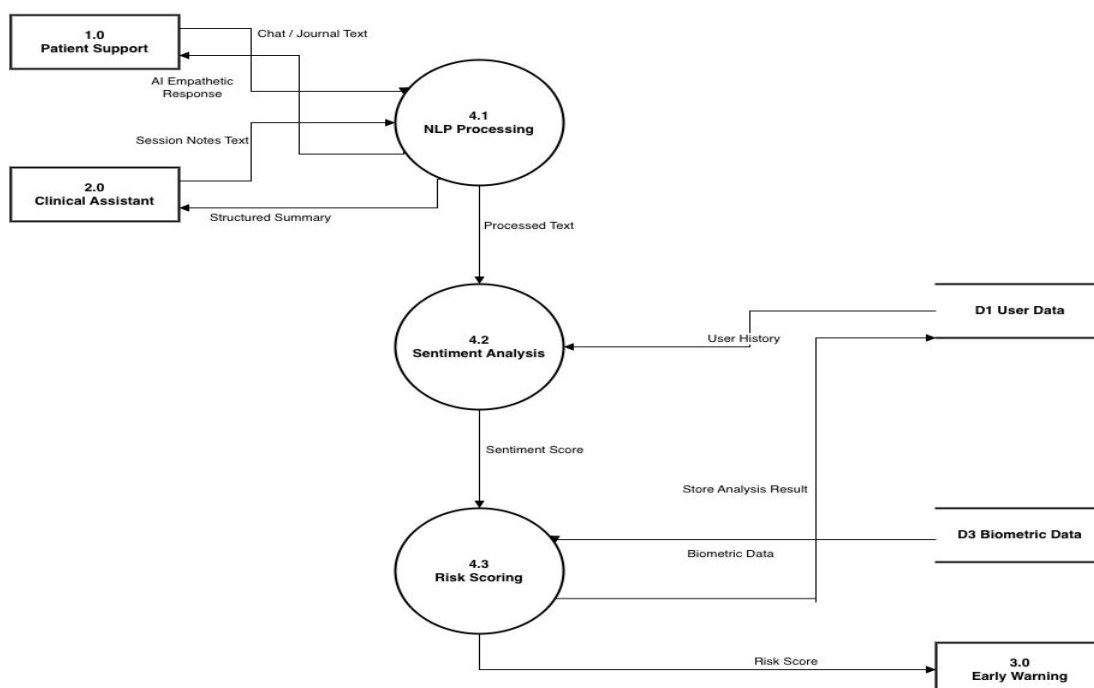
2.0 Clinical Assistant



3.0 Early Warning System



4.0 AI/ML Processing



5.0 User Connectivity

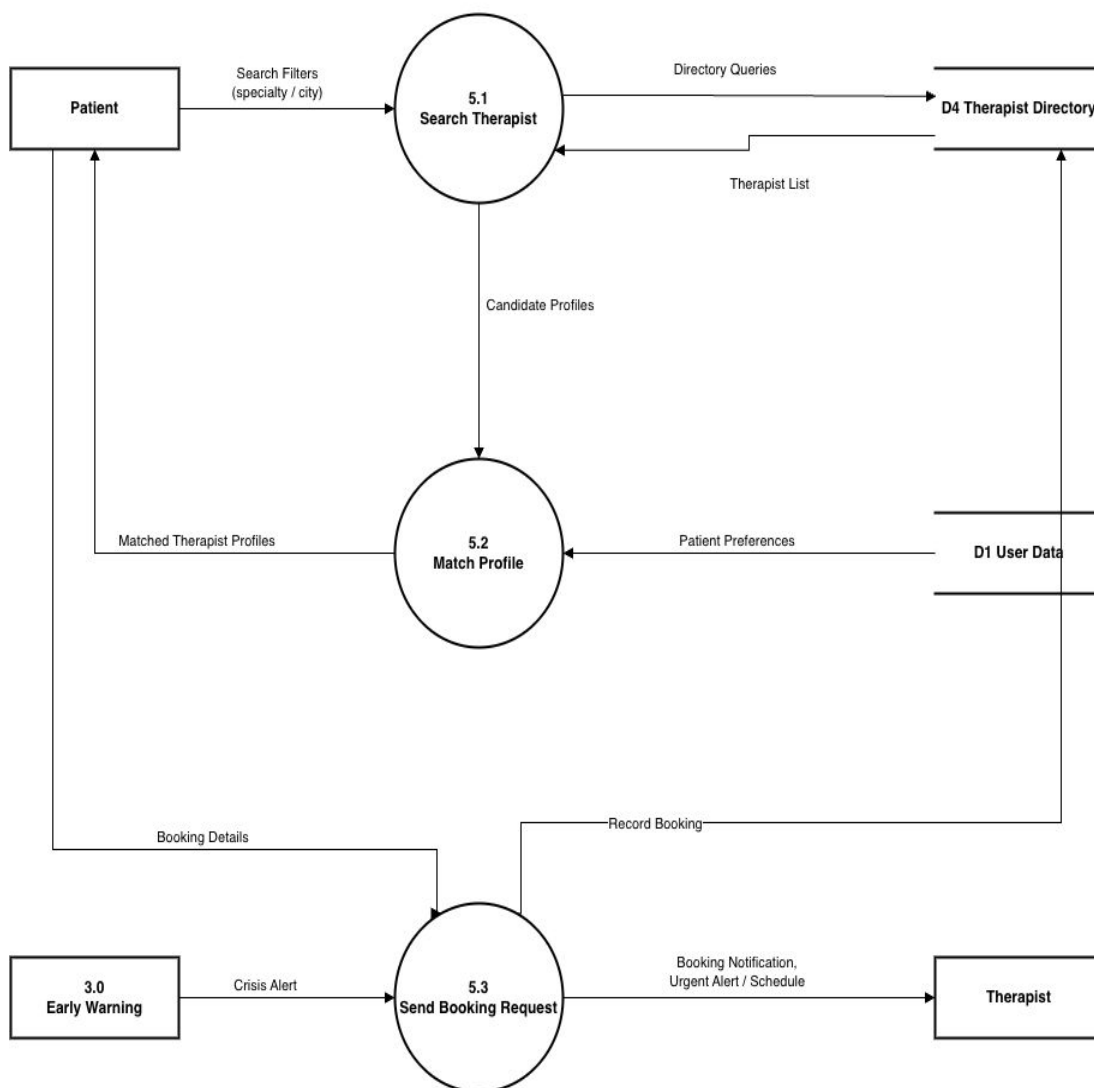


Figure 17: DFD Level 2

11 Domain Model

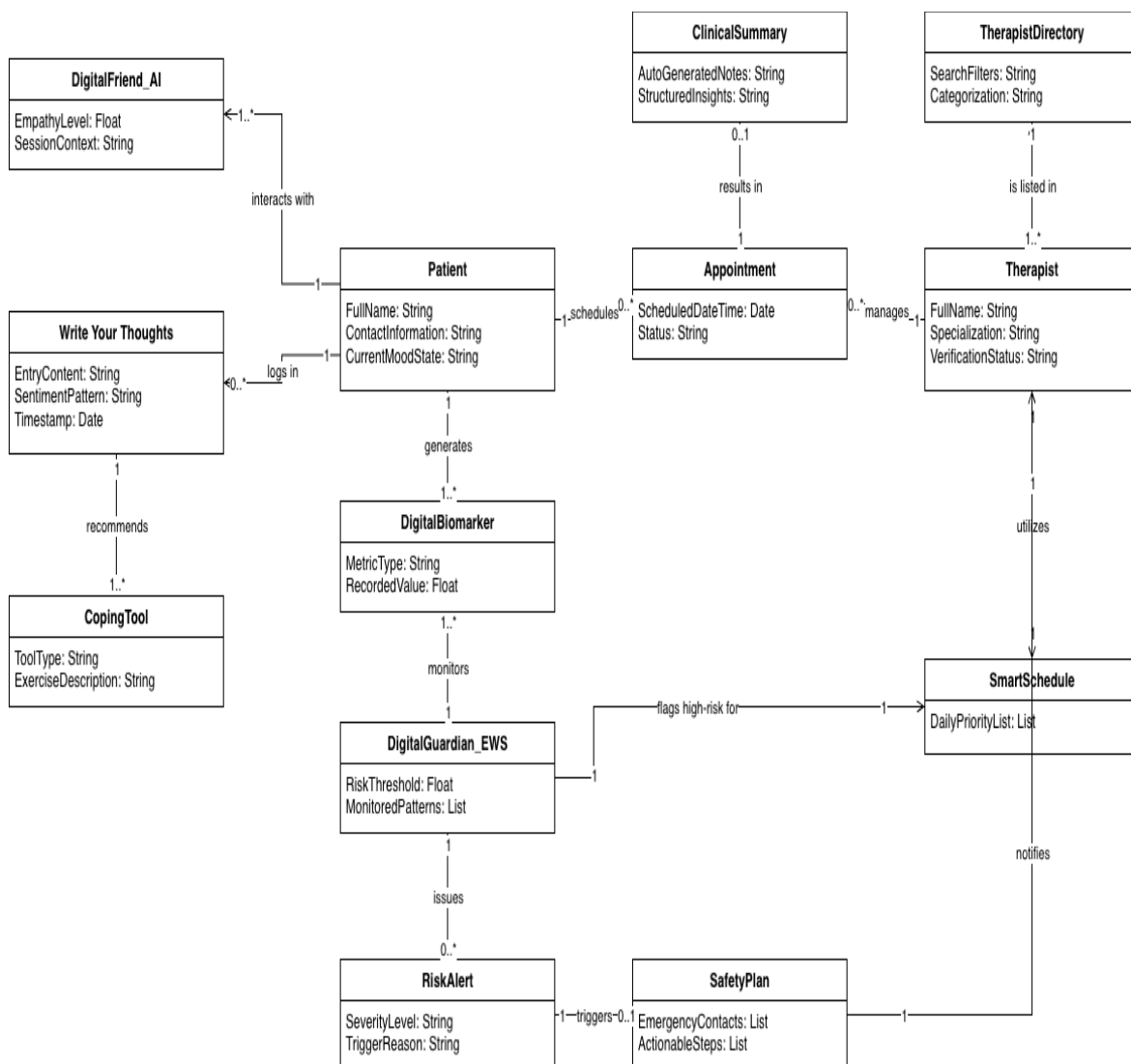


Figure 18: Domain Model

12 Class Diagram

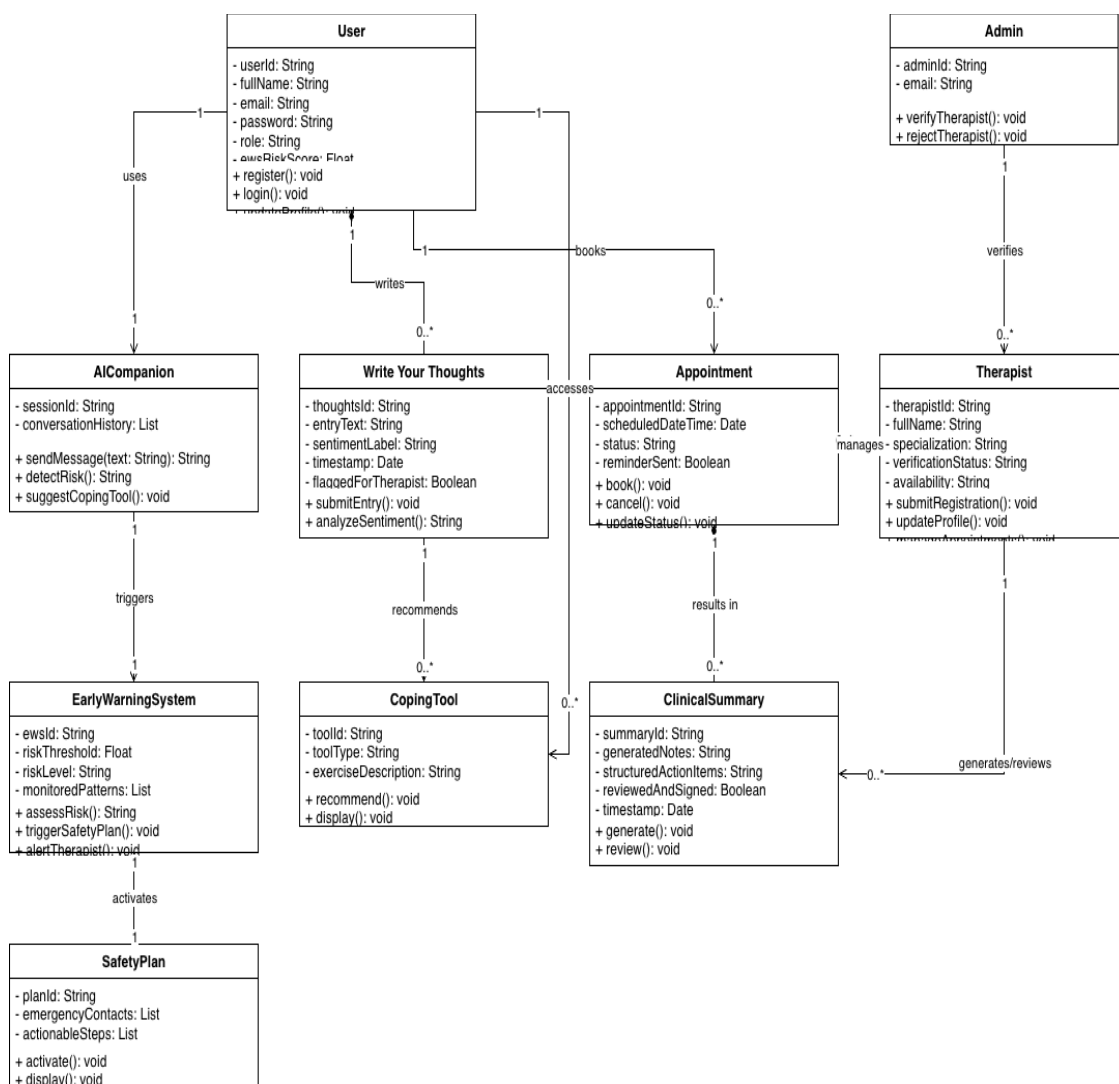


Figure 19: Class Diagram

13 Package Diagram

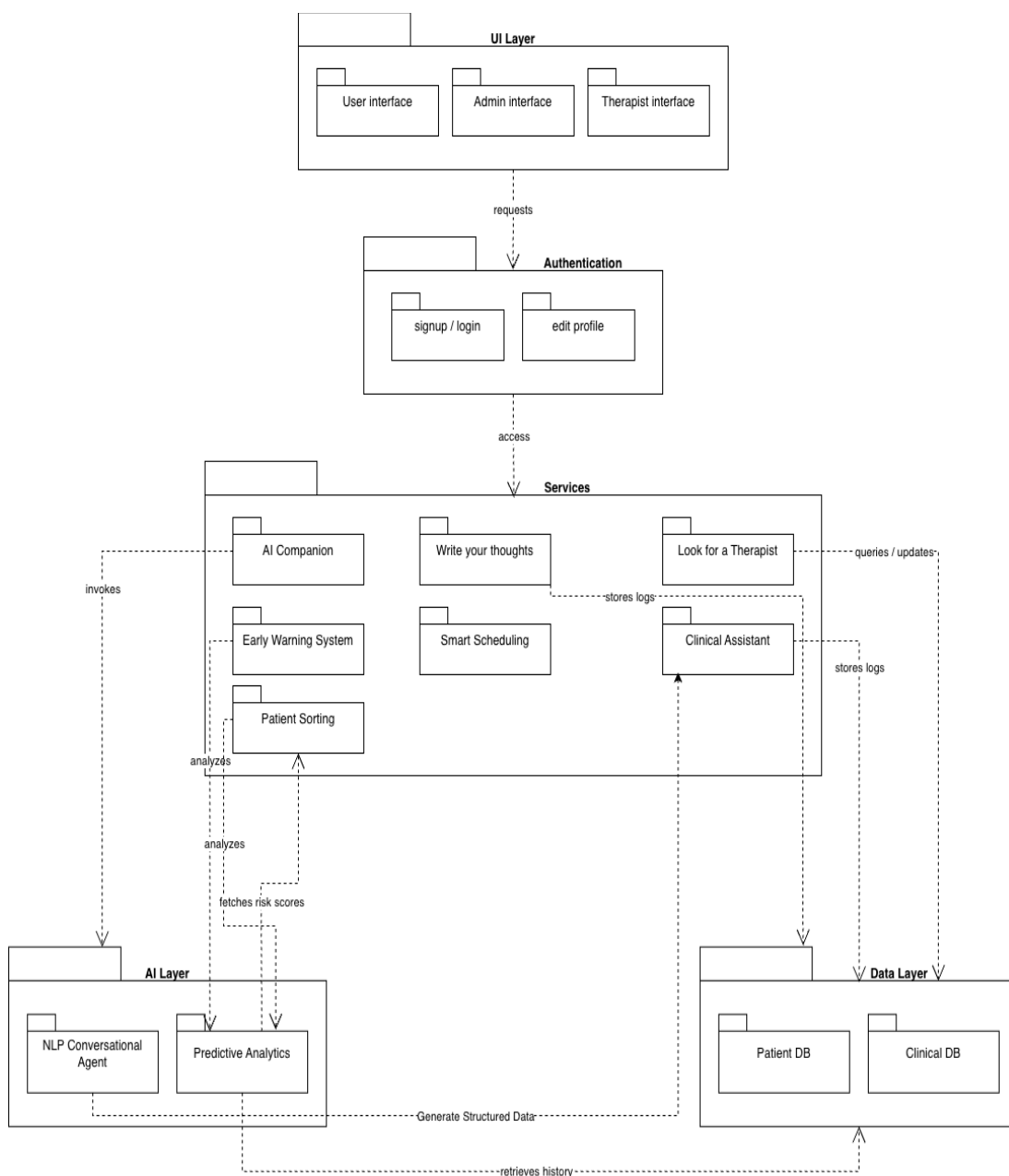


Figure 20: Package Diagram

14 System Data Architecture

14.1 Schema Design

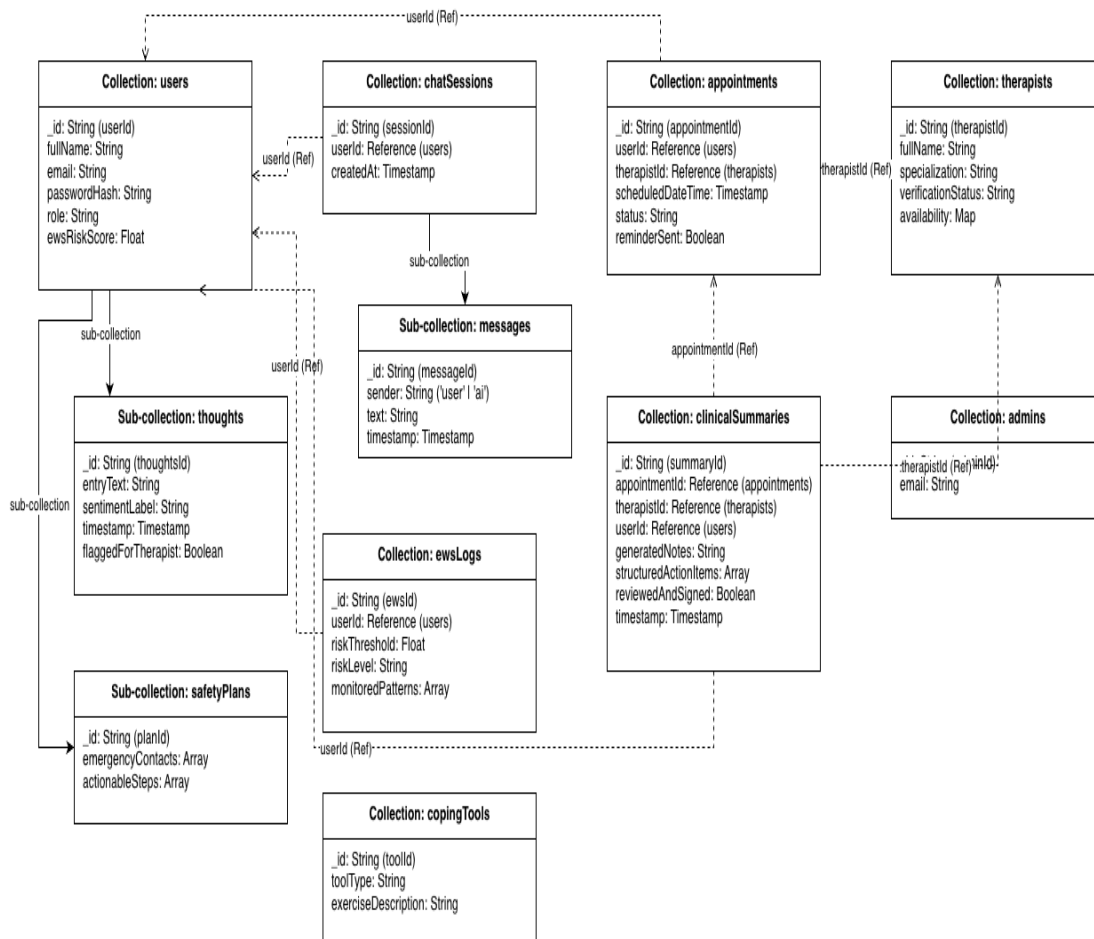


Figure 21: Schema Design

15 Architecture Diagram

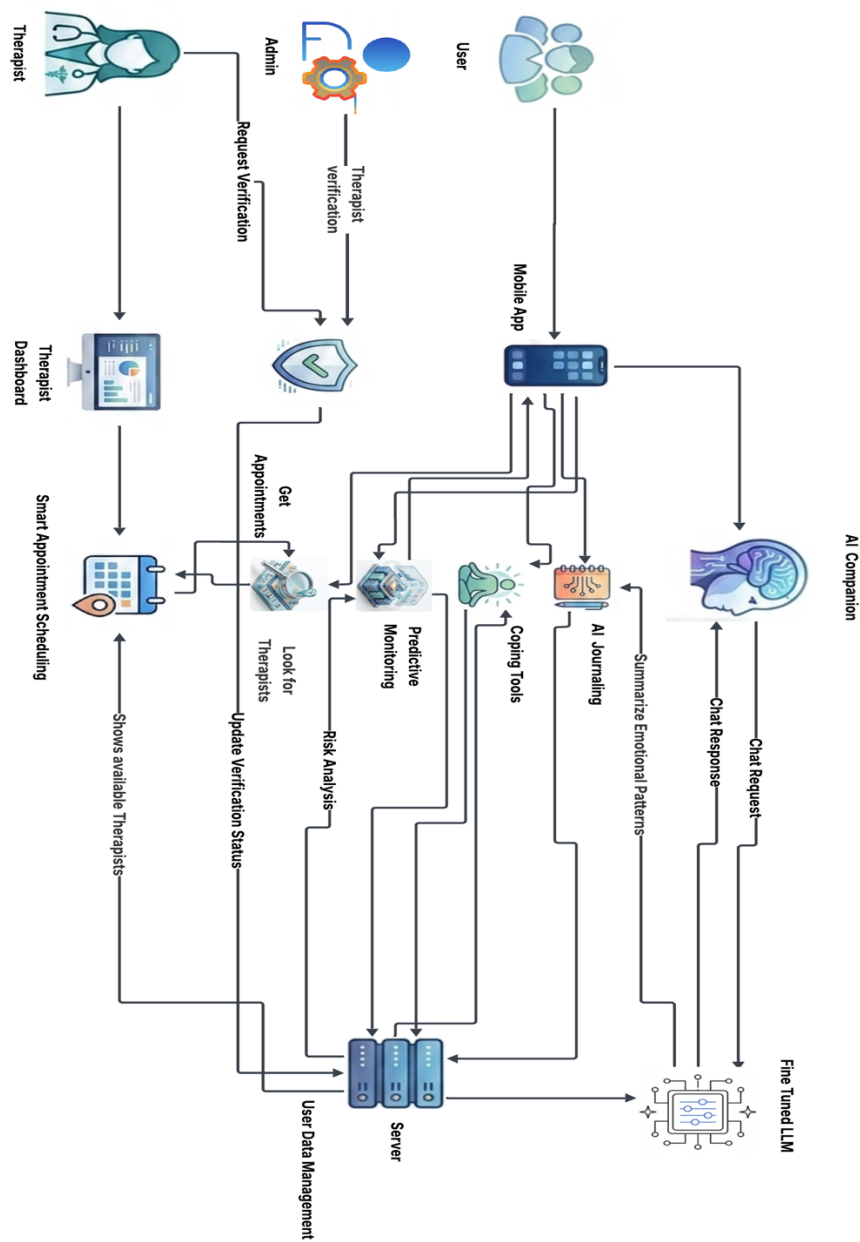


Figure 22: Architecture Diagram

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